REAPER-Notator v2.50-beta.3

Comprehensive Codebase Volume & Structural Analysis

TOTAL LINES	EXECUTABLE CODE	SOURCE FILES	PATTERN LIBRARY
75,124	64,023	89 Lua	1,500 JSON

1. Architecture & Module Volume Breakdown

Module Directory	Files	Total Lines	Code Lines	Comments	Blank	Code Ratio
modules/services	35	39,230	33,673	2,368	3,361	85.8%
modules/rendering	7	14,018	12,032	849	1,187	85.8%
modules/ui	24	12,345	10,089	891	1,391	81.7%
tests	2	2,726	2,444	116	181	89.7%
modules	3	2,621	2,308	472	161	88.1%
modules/input	2	2,073	1,807	92	176	87.2%
Root	1	1,321	1,057	161	107	80.0%
modules/classes	15	790	613	110	96	77.6%
TOTAL	**89**	**75,124**	**64,023**	**5,059**	**6,660**	**85.2%**

2. Key Core System Files by Volume (Top 15)

File Name	Subsystem / Module	Lines	Code	Comments
midi_service.lua	modules/services	7,467	6,592	288
score_canvas.lua	modules/rendering	5,105	4,456	244
engraver.lua	modules/rendering	4,199	3,618	303
slur_service.lua	modules/services	2,654	2,314	125
record_quantize_service.lua	modules/services	2,500	2,167	162
canvas_context_menus.lua	modules/rendering	2,339	2,045	105
musicxml_import_service.lua	modules/services	2,193	1,911	125
musicxml_export_service.lua	modules/services	1,715	1,452	145
melody_pathfinder_service.lua	modules/services	1,571	1,313	113
portamento_service.lua	modules/services	1,561	1,322	90
glissando_service.lua	modules/services	1,489	1,257	82
midi_editor_drawer.lua	modules/ui	1,449	1,206	118
test_all_modules.lua	tests	1,447	1,232	90
settings_modal.lua	modules/ui	1,420	1,154	96
state.lua	modules	1,415	1,237	236

3. Curated Orchestral Pattern Library (1,500 Patterns)

Category Folder	Genre / Instrumentation Scope	Pattern Count
01_Strings_Staccato	Cinematic Strings (Staccato / Spiccato Ostinatos)	150 patterns
02_Strings_Pizzicato	Orchestral Strings (Pizzicato / Bartók Snap)	150 patterns
03_Brass_Blockbuster	Epic Brass (Horns, Trumpets, Trombones, Tubas)	150 patterns
04_Cinematic_Melodies	Heroic Themes & Symphonic Lead Melodies	150 patterns
05_Counter_Melodies	Polyphonic Counter-Melodies & Descant Voices	150 patterns
06_Woodwinds_Textures	Woodwind Runs, Flutter-Tongue & Textures	150 patterns
07_Cinematic_Piano	Concert Grand Piano (Arpeggios & Ostinatos)	150 patterns
08_Ancient_Harp_Greek_Roman	Antiquity Concert Harp (Greek & Roman Hymns)	150 patterns
09_Orchestral_Percussion	Orchestral & Cinematic Percussion (Timpani, Taiko, Drums)	150 patterns
10_Low_Strings_Celli_Basses	Low Strings Tension & Rhythm (Staccato Celli & Double Basses)	150 patterns
TOTAL PATTERNS	**10 Curated Production Categories**	**1,500 patterns**

4. Specialized Orchestral & Engraving Services Subsystem

Service Subsystem File	Musical & Architectural Responsibility	Code Lines
annotation_service.lua	Annotation Layer Engine - Per-track multi-layer free	972
melody_pathfinder_service.lua	Melodic Pathfinder Engine - Corpus-trained Viterbi i	1,313
melody_style_corpus.lua	Composer Style Corpus - Statistical Markov transitio	248
divisi_service.lua	Divisi Engine - Split polyphonic voices onto full du	434
divisio_vocum_service.lua	Divisio Vocum - Orchestral Voice Division across tra	308
dynamics_engine.lua	Orchestral Dynamics Engine - Bezier/Linear CC automa	1,128
filter_notes_service.lua	Filter Irregular Notes - Metric onset analysis and v	308
frame_timer_service.lua	Native High-Refresh Timer - 1ms kernel multimedia ti	147
glissando_service.lua	Glissando Engine - Smooth chromatic / diatonic slide	1,257
midi_service.lua	Core MIDI Engine - Low-latency take I/O and event sy	6,592
musicxml_export_service.lua	MusicXML 4.0 Exporter - Standard score interchange f	1,452
musicxml_import_service.lua	MusicXML 4.0 Importer - Orchestral part loading and	1,911
audio_preview.lua	Audition Preview Engine - Non-destructive note click	951
reaticulate_parser.lua	Reaticulate Bridge - FX Slot 1 setup, bank sync & ke	876
record_quantize_service.lua	Record & Auto-Quantize Engine - Live MIDI capture &	2,167
scale_service.lua	Scale & Harmony Service - Chord Track harmonic trans	835
slur_service.lua	Slur Engine - Ergonomic Bezier phrasing curves and k	2,314
portamento_service.lua	Portamento Service - High-resolution micro-tonal pit	1,322
pattern_service.lua	Pattern Engine - Orchestral motif browser and timeli	582
rehearsal_mark_service.lua	Rehearsal Marks - Custom/auto structural navigation	368
fermata_service.lua	Fermata Service - Score-wide pause holds and tempo m	232
pedal_service.lua	Pedal Service - Classical damper and sustain pedal C	530

Quality Assurance & Standards Status: 100% Certified

All 89 Lua files, 152 unit tests, and 1,500 pattern JSON files verified with 0 errors. Fully compliant with Gardner Read & Elaine Gould engraving standards.